Final Assessment — Introduction to Data Science

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```r
knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE)

## ----------------------------------------------------------------------------
##  DO NOT EDIT personalise.R. Run this chunk once to build your dataset.
## ----------------------------------------------------------------------------
source("personalise.R")

my <- make_my_dataset(params$studentID,
                      raw_dir = "data_raw",
                      out_dir = "data")
```

```
## Student ID   : u3284342
## Seed         : 12109058
## Window       : 2004 - 2023
## Countries    : 150
## Files written to /home/ruhanshafi/Projects/IntroDS/Assessment 2/data
##   Emissions.csv   3005 rows
##   Renewables.csv  148 rows x 21 cols
##   Consumption.csv 1897 rows
##   Countries.csv   150 rows
## FINGERPRINT  : EM-606669-0798-6584
```

```r
# Load the packages you need here.
# library(tidyverse)
# library(lubridate)
```

Introduction

Briefly introduce the dataset, the questions you set out to answer, and how this report is organised. Half a page is plenty.

YOUR ANSWER

Assumptions

State any assumptions you are making about the data or the analysis before you begin.

YOUR ANSWER

Task 1: Data preparation and wrangling

1.1 Import the four CSV files

1.2 Reshape `Renewables.csv` to one row per country per year

Remember: this file contains rows that are not countries.

What did you do with the non-country rows, and why?

YOUR ANSWER

1.3 Rename the columns

1.4 Make `Year` consistent across all data frames

1.5 Merge the five new variables into the master data frame

Add `Renewables`, `ConsCO2`, `Population`, `GDP`, `GDPCapita`.

How many rows failed to match on each join, and what did you do about them?

YOUR ANSWER

1.6 Handle the NAs

Justify your treatment of each variable. Different variables may warrant different choices.

YOUR ANSWER

1.7 Add `Decade` and `Period`

1.8 Add the four cumulative variables

`CumCO2`, `CumOilCO2`, `CumRenew`, `CumConsCO2`.

1.9 Add `NetImported` and `OilShare`

1.10 Add the four per-capita variables

`CO2_PC`, `OilCO2_PC`, `Renew_PC`, `ConsCO2_PC`.

```
# A quick look at your finished master data frame.
# dim(master)
# str(master)
```

1A. Data Quality Audit Report

Approximately 250 words in total across the three parts below. Every claim must reference a specific value, country, year or column from ***your*** *data.*

i. Defect inventory

Complete the table. Add rows as needed — there are more defects than there are rows here.

File	Column	Rows affected	How I detected it	How I handled it

```
# The code you used to find the defects.
```

ii. The negative emissions problem

```
# Identify the negative-emissions rows by country and year.
```

Which rows are they? Why do negative values legitimately appear in published emissions series? What did you do, and what alternative did you reject?

YOUR ANSWER

iii. Coverage gap in consumption-based data

```
# Count the countries with no consumption-based data, and compare the
# countries that do and do not have coverage.
```

How many countries have no coverage? Name three. Is imputing zero valid? If it biases `ConsCO2_PC` *and* `NetImported`*, in which direction, and for which type of country is the bias worst? Support this with a number.*

YOUR ANSWER

Task 2: Exploratory data analysis

2.1 Highest-emitting year, and the largest year-on-year fall

What caused the largest fall?

YOUR ANSWER

2.2 Cumulative totals over time for the whole dataset

2.3 Extract `lastYear_data`

Compute the final year — do not type it in.

2.4 Top 10 countries on four measures

`top10total`, `top10percap`, `top10net`, `top10renew`.

2.5 Totals by region

2.6 Cumulative emissions over time, top 10 countries (log Y)

2.7 Faceted plot, one panel per country

2.8 Top 10 by emissions per capita

2.9 Total, average and median cumulative emissions by region

2.10 Top 2 countries per region by emissions per capita

2A. Counterintuitive finding

What did you expect, and why was that reasonable?

YOUR ANSWER

What does the data actually show? Quote a specific value, country or trend from your own plots.

YOUR ANSWER

What real-world mechanism could explain the discrepancy?

YOUR ANSWER

Supporting visualisation

Caption (2–3 sentences): what does this chart show, and why did you choose this chart type?

YOUR ANSWER

2B. Visualisation design critique

Visualisation chosen: *(state which question it is from)*

Approximately 150 words covering: two design decisions and what each communicates; one alternative you rejected and concretely why it would communicate worse; one limitation a reader could misinterpret, and how you would mitigate it.

YOUR ANSWER

Task 3: Data-driven modelling

3.1 Build `cor_data`

3.2 Correlation matrix

Report `cor(CumCO2, GDP)` and `cor(CumCO2, GDPCapita)`. Why are those two numbers so different?

YOUR ANSWER

3.3 Distribution of `CumCO2`, with and without log

3.4 Train/test split

```
## Required - do not remove or change this line:
set.seed(seed_from_id(params$studentID))

# Your 65/35 split goes below.
```

3.5 Linear model: `CumCO2 ~ GDP`

3.6 Linear model: all predictors

3.7 Interpret and compare the two models

When should the first be preferred, and when the second? If the full model has the higher test RMSE, explain why — that is not an error.

YOUR ANSWER

3.8 Alternative models

Comparison table:

Model	RMSE	R-squared
Linear (GDP only)		
Linear (all variables)		

YOUR ANSWER *(brief comment on the comparison)*

3A. Assumption checking

i. Linearity

Does a linear relationship look reasonable? Which predictor departs from it most clearly?

YOUR ANSWER

ii. Distribution of the response

Using your plots from 3.3 — does raw `CumCO2` satisfy the normality assumption? Is a transformation warranted?

YOUR ANSWER

iii. Residual diagnostics

Plot	What assumption it tests	Does your model satisfy it?
Residuals vs Fitted		
Normal Q-Q		
Scale-Location		
Residuals vs Leverage		

iv. Influential observations

```
# Identify the high-leverage country, then refit without it.
```

High-leverage country: *name it*

New test RMSE without it: *value*

Is removing it defensible? Two sentences.

YOUR ANSWER

v. Verdict

Is linear regression a valid choice here? If assumptions are violated, what does that mean in practice for your coefficients and predictions?

YOUR ANSWER

3B. Feature engineering critique

Approximately 300 words across the three parts.

i. Theoretical critique

State the physical pathway from GDP to emissions. Then explain why high correlation still does not license a causal claim: name at least two confounding or mediating variables GDP is proxying for. Reference your own correlation from 3.2, and contrast it with the `GDPCapita` correlation.

YOUR ANSWER

ii. Proposed feature

Propose one engineered feature not in the dataset. Say exactly how you would construct it — what raw data, what calculation — and why it is more causally connected to emissions than GDP.

YOUR ANSWER

iii. Ethical risk

If this model allocated climate finance or set national reduction obligations, name one specific way it would be systematically unfair, and show that the unfairness comes from the structure of the model, not just "bias in the data".

Use your own data: name a country with large positive `NetImported` *and one with large negative* `NetImported`*, and build your argument around them.*

```
# Pull out the two countries you want to use.
```

YOUR ANSWER

Task 4: Insights

Narrative thesis

One sentence: the single story a reader should take away from your dashboard.

YOUR ANSWER

Dashboard sketch

Embed your photographed hand-drawn sketch below. This is mandatory — without it you lose all design marks.

4.1 Data augmentation

How would you augment this dataset for a comprehensive view of the transition? What sources, and what would each add?

YOUR ANSWER

4.2 Key insights

Which insights would you highlight for a public audience? What figures would you use, and why is each important?

YOUR ANSWER

4.3 Design elements

Layout, colour, typography, placement.

YOUR ANSWER

4.4 Audience consideration

How does the design serve a public audience — accessible, understandable, actionable?

YOUR ANSWER

4.5 Additional questions

One relevant metric or visualisation you deliberately omitted, and why omitting it serves clarity.

YOUR ANSWER

One way your dashboard could be misread, and how design mitigates it.

YOUR ANSWER

Does your headline figure use total or per-capita emissions? Defend that choice against the obvious objection.

YOUR ANSWER

Conclusions

What are the main things your analysis shows? Keep it grounded in your own numbers.

YOUR ANSWER

Section 5: Verification

5.1 Results Summary Table

Fill every cell by hand from your own output. If you could not compute one, write `NA` and say why in a footnote below the table.

#	Quantity	Your value
1	Dataset fingerprint (printed by the setup chunk)	
2	First and last year in your dataset	
3	Number of countries in your `Emissions.csv`	
4	`dim()` of your final merged master data frame	
5	Number of duplicate rows you removed in Task 1	
6	Countries and years of the negative-emissions rows (1A ii)	
7	Number of countries with no consumption-based data (1A iii)	
8	Year with the highest total CO2, and that total (Task 2.1)	
9	Top country by cumulative emissions, and by emissions per capita	
10	`cor(CumCO2, GDP)` and `cor(CumCO2, GDPCapita)` (2 dp)	
11	Test RMSE of the GDP-only model (Task 3.5)	
12	Test RMSE of the full model (Task 3.6)	
13	The high-leverage country you identified (3A iv)	

5.2 Reproducibility appendix

Run this chunk unchanged. Do not edit it.

```
cat("Student ID  :", params$studentID, "\n")
```

```
## Student ID  : u3284342
```

```
cat("Fingerprint :", my$fingerprint, "\n")
```

```
## Fingerprint : EM-606669-0798-6584
```

```
cat("Window      :", paste(my$window, collapse = " - "), "\n")
```

```
## Window      : 2004 - 2023
```

```r
cat("Knitted at  :", format(Sys.time(), "%Y-%m-%d %H:%M:%S %Z"), "\n")
```

```
## Knitted at  : 2026-11-09 18:06:52 AEDT
```

```r
cat("Chunks run  :", length(knitr::all_labels()), "\n\n")
```

```
## Chunks run  : 40
```

```r
print(sessionInfo()$R.version$version.string)
```

```
## [1] "R version 4.6.1 (2026-06-24)"
```

Appendix: Decision Log

*Minimum five entries. Write each one **at the time you make the decision**, not at the end. Every entry must reference something specific from your own analysis — a variable name, a country, a line of code, an output value. Generic entries receive zero marks. Add rows freely; more than five is viewed favourably.*

#	Task & step	Decision made	Alternative considered	Why I chose this	Impact on results
1					
2					
3					
4					
5					

Declaration

I confirm that the dataset analysed in this submission was generated from my own student ID, that the values reported in Section 5.1 are the values produced by the code in this document, and that all explanations and interpretations are my own work.

Student ID: _____ **Date:** _____